

# Preflight Readiness Review

Create and conduct the most reliable preflight inspection for a classroom rover or mechanism.

|          |                   |             |
|----------|-------------------|-------------|
| Advanced | 1-2 class periods | Competition |
|----------|-------------------|-------------|

## Engineering Context

Two student teams are preparing identical systems for a mission demonstration. The winning team is not simply the fastest; it is the team whose checklist catches the most faults while remaining efficient.

## What You Will Practice

- Translate failure risks into inspection steps
- Sequence checks efficiently
- Use objective pass/fail criteria
- Validate a checklist through fault insertion

## What You Need

- Classroom rover or mechanism
- Fault cards or instructor-inserted issues
- Checklist template
- Timer
- Engineering notebook

## Student Instructions

- 1 List at least ten possible mission-preventing faults.
- 2 Rank each fault by likelihood and consequence.
- 3 Write a checklist with observable pass/fail criteria.
- 4 Arrange checks from safe/no-power inspection through powered functional checks.
- 5 Exchange systems with another team; the instructor inserts approved faults.
- 6 Conduct the inspection while being timed.
- 7 Record detected faults, missed faults, and unnecessary checks.
- 8 Revise the checklist and repeat one validation round.

## Engineering Requirements

- At least 12 checklist items
- Checklist must include identity, physical condition, power, movement, sensor/control, and final configuration
- No item may use vague wording such as “looks good”
- Score balances detection rate and completion time

## Constraints & Safety

- Only instructor-approved faults may be inserted
- Do not defeat guards or create electrical hazards
- Power must remain off during physical inspection unless instructed

## What You Submit

- Risk-ranked fault list
- Preflight checklist
- Validation results
- Revised checklist
- Competition score summary

## Reflection Questions

- 1 Which fault was easiest to miss?
- 2 Which checklist item had the clearest pass/fail standard?
- 3 What check could be automated?
- 4 How did time pressure affect performance?

|                                                                                                                                     |                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Tip</b><br>Write checklist items as actions with a condition: “Verify battery connector is fully seated and locked.” | <b>Common Mistake</b><br>A long checklist is not automatically better. Repeated or low-value checks can hide critical items.                                |
| <b>Stretch Goal</b><br>Convert the final checklist into a QR-linked digital form.                                                   | <b>Career Connection</b><br>Aerospace operations teams use configuration-controlled checklists for aircraft, launch vehicles, and ground support equipment. |